

Chapter 2: Cloud Architecture

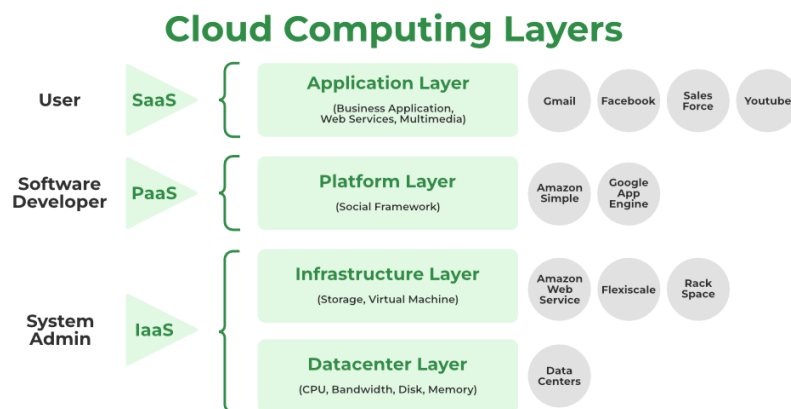
Layers and Models Layers in cloud architecture:

It is possible to organize all the concrete realizations of cloud computing into a layered view covering the entire, from hardware appliances to software systems.

All of the physical manifestations of cloud computing can be arranged into a layered picture that encompasses anything from software systems to hardware appliances. Utilizing cloud resources can provide the "computer horsepower" needed to deliver services. This layer is frequently done utilizing a data center with dozens or even millions of stacked nodes. Because it can be constructed from a range of resources, including clusters and even networked PCs, cloud infrastructure can be heterogeneous in character. The infrastructure can also include database systems and other storage services.

The core middleware, whose goals are to create an optimal runtime environment for applications and to best utilize resources, manages the physical infrastructure. Virtualization technologies are employed at the bottom of the stack to ensure runtime environment modification, application isolation, sandboxing, and service quality. At this level, hardware virtualization is most frequently utilized. The distributed infrastructure is exposed as a collection of virtual computers via hypervisors, which control the pool of available resources. By adopting virtual machine technology, it is feasible to precisely divide up hardware resources like CPU and memory as well as virtualize particular devices to accommodate user and application needs.

Layered Architecture of Cloud



Application Layer

1. The application layer, which is at the top of the stack, is where the actual cloud apps are located. Cloud applications, as opposed to traditional applications, can take advantage of the automatic-scaling functionality to gain greater performance, availability, and lower operational costs.
2. This layer consists of different Cloud Services which are used by cloud users. Users can access these applications according to their needs. Applications are divided into Execution layers and Application layers.

3. In order for an application to transfer data, the application layer determines whether communication partners are available. Whether enough cloud resources are accessible for the required communication is decided at the application layer. Applications must cooperate in order to communicate, and an application layer is in charge of this.
4. The application layer, in particular, is responsible for processing IP traffic handling protocols like Telnet and FTP. Other examples of application layer systems include web browsers, SNMP protocols, HTTP protocols, or HTTPS, which is HTTP's successor protocol.

Platform Layer

1. The operating system and application software make up this layer.
2. Users should be able to rely on the platform to provide them with Scalability, Dependability, and Security Protection which gives users a space to create their apps, test operational processes, and keep track of execution outcomes and performance. SaaS application implementation's application layer foundation.
3. The objective of this layer is to deploy applications directly on virtual machines.
4. Operating systems and application frameworks make up the platform layer, which is built on top of the infrastructure layer. The platform layer's goal is to lessen the difficulty of deploying programmers directly into VM containers.
5. By way of illustration, Google App Engine functions at the platform layer to provide API support for implementing storage, databases, and business logic of ordinary web apps.

Infrastructure Layer

1. It is a layer of virtualization where physical resources are divided into a collection of virtual resources using virtualization technologies like Xen, KVM, and VMware.
2. This layer serves as the Central Hub of the Cloud Environment, where resources are constantly added utilizing a variety of virtualization techniques.
3. A base upon which to create the platform layer. constructed using the virtualized network, storage, and computing resources. Give users the flexibility they want.
4. Automated resource provisioning is made possible by virtualization, which also improves infrastructure management.
5. The infrastructure layer sometimes referred to as the virtualization layer, partitions the physical resources using virtualization technologies like Xen, KVM, Hyper-V, and VMware to create a pool of compute and storage resources.
6. The infrastructure layer is crucial to cloud computing since virtualization technologies are the only ones that can provide many vital capabilities, like dynamic resource assignment.

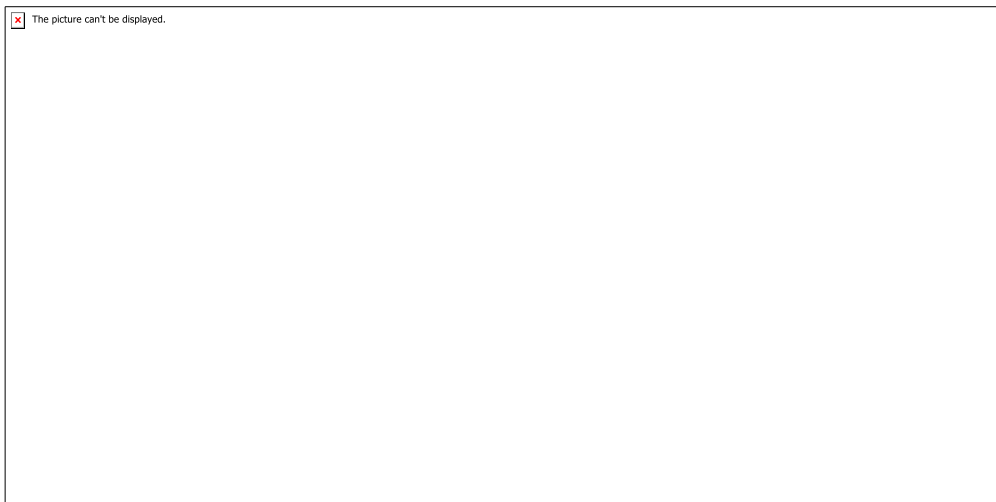
Datacenter Layer

- In a cloud environment, this layer is responsible for Managing Physical Resources such as servers, switches, routers, power supplies, and cooling systems.
- Providing end users with services requires all resources to be available and managed in data centers.
- Physical servers connect through high-speed devices such as routers and switches to the data center.
- In software application designs, the division of business logic from the persistent data it manipulates is well-established. This is due to the fact that the same data cannot be incorporated into a single application because it can be used in numerous ways to support numerous use cases. The requirement for this data to become a service has arisen with the introduction of microservices.
- A single database used by many microservices creates a very close coupling. As a result, it is hard to deploy new or emerging services separately if such services need database modifications that may have an impact on other services. A data layer containing many databases, each serving a single microservice or perhaps a few closely related microservices, is needed to break complex service interdependencies.

SaaS, PaaS and IaaS

Cloud computing provides on-demand IT resources that can easily scale to meet business demands. Rather than maintaining complex physical hardware and software on-premises, organizations can rent these services over the internet.

The cloud ecosystem is divided into three primary service models: IaaS, PaaS, and SaaS. Each model offers a different level of control, flexibility, and management, catering to different business needs from raw infrastructure to ready-to-use software



1. Infrastructure as a Service (IaaS)

IaaS provides virtualized computing resources such as servers, storage, and networking over the internet on a pay-as-you-go basis. It offers the highest level of control and flexibility, acting as the digital foundation for your IT operations.

How it works: It is like renting an empty plot of land. The provider gives you the space and the utilities (hardware, network, storage), but you are entirely responsible for building the house (installing the OS, middleware, and applications).

Characteristics:

- Eliminates the capital expense of buying physical servers.
- Offers complete control over operating systems, applications, and development frameworks.
- Resources can be scaled up or down instantly based on traffic spikes.

Real-World Use Case: A tech startup experiencing rapid growth can use IaaS to instantly increase their server capacity to handle web traffic without needing to purchase, house, or wire expensive physical hardware.

Target Audience: IT Administrators, Network Architects, and DevOps teams.

Popular Providers: Amazon Web Services (AWS EC2), Microsoft Azure (Virtual Machines), Google Compute Engine (GCE), DigitalOcean.

2. Platform as a Service (PaaS)

PaaS offers a complete, managed cloud environment specifically designed for developing, testing, running, and managing applications. It abstracts the underlying infrastructure so developers can focus purely on writing code.

How it works: It is like renting an unfurnished house. The foundation, plumbing, and electricity are already built and maintained by the landlord; you just need to bring your furniture and decorate (write and deploy your code).

Characteristics:

- Acts as a complete toolkit for developers, providing pre-built tools, libraries, and development environments.
- The cloud provider handles all backend infrastructure, including server provisioning, OS patching, and network routing.
- Automates deployment workflows and allows for easy team collaboration.

Real-World Use Case: A software development team building a mobile app can use PaaS to write their code and automatically connect it to managed databases and authentication services without ever having to configure a backend server.

Target Audience: Software Developers and Application Delivery Teams.

Popular Providers: AWS Elastic Beanstalk, Google App Engine, Heroku, Microsoft Azure App Service.

3. Software as a Service (SaaS)

SaaS is the most user-friendly and widely used cloud model. It delivers fully functional, ready-to-use software applications over the internet, typically on a subscription basis.

How it works: It is like staying in a fully furnished, serviced hotel room. Everything is provided, maintained, and cleaned for you; you simply walk in and use it.

Characteristics:

- Applications are ready to use immediately directly through a web browser or mobile app.
- The cloud provider handles absolutely everything: infrastructure, software updates, bug fixes, and security maintenance.
- Requires minimal technical expertise to operate.

Real-World Use Case: A global sales team uses a cloud-based CRM to track leads and communicate with clients. They simply log in via their browsers; they don't need to worry about where the data is hosted or how the software is updated.

Target Audience: End-users, Business Professionals, and Consumers.

Popular Providers: Salesforce, Google Workspace (Gmail, Docs), Microsoft 365, Zoom, Slack.

Difference Between SaaS, PaaS, and IaaS

Basis Of	IaaS	PaaS	SaaS
Access	IaaS gives access to the resources like virtual machines and virtual storage.	PaaS gives access to run time environment to deployment and development tools for application.	SaaS gives access to the end user.
Model	It is a service model that provides virtualized computing resources over the internet.	It is a cloud computing model that delivers tools that are used for the development of applications.	It is a service model in cloud computing that hosts software to make it available to clients.
Technical understanding.	It requires technical knowledge.	Some knowledge is required for the basic setup.	There is no requirement about technicalities company handles everything.

Basis Of	IaaS	PaaS	SaaS
Popularity	It is popular among developers and researchers.	It is popular among developers who focus on the development of apps and scripts.	It is popular among consumers and companies, such as file sharing, email, and networking.
Percentage rise	It has around a 12% increment.	It has around 32% increment.	It has about a 27 % rise in the cloud computing model.
Usage	Used by the skilled developer to develop unique applications.	Used by mid-level developers to build applications.	Used among the users of entertainment.
Cloud services.	Amazon Web Services, sun, vCloud Express.	Facebook, and Google search engine.	MS Office web, Facebook and Google Apps.
Enterprise services.	AWS virtual private cloud.	Microsoft Azure.	IBM cloud analysis.
Outsourced cloud services.	Salesforce	Force.com, Gigaspace.	AWS, Terremark
User Controls	Operating System, Runtime, Middleware, and Application data	Data of the application	Nothing
Others	It is highly scalable and flexible.	It is highly scalable to suit the different businesses according to resources.	It is highly scalable to suit the small, mid and enterprise level business

Cloud deployment models:

Cloud computing has become an essential part of modern business, offering unparalleled flexibility, scalability, and cost-effective solutions. However, to harness its full potential, selecting the most appropriate cloud deployment model is critical. Whether you are a small startup or a massive enterprise, your choice will directly dictate your organization's security, scalability, and operational efficiency

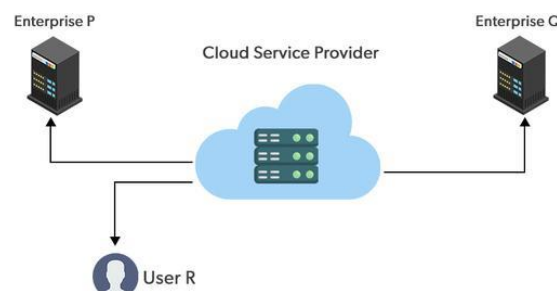
What is a Cloud Deployment Model?

A cloud deployment model defines where your cloud infrastructure lives, who owns and manages it, and how it is accessed. It establishes the boundaries of your cloud environment, dictating what you can customize, how resources are shared among users, and the overarching purpose of the setup.

Understanding these models is the crucial first step for any business migrating to the cloud, as each offers distinct trade-offs in governance, cost, security, and management.

Public Cloud

The public cloud delivers infrastructure and services over the internet to the general public or broad industry groups. The infrastructure is entirely owned, managed, and maintained by a third-party cloud service provider (e.g., Google Cloud, AWS, Microsoft Azure). Resources are shared among multiple tenants.



Advantages of the Public Cloud Model

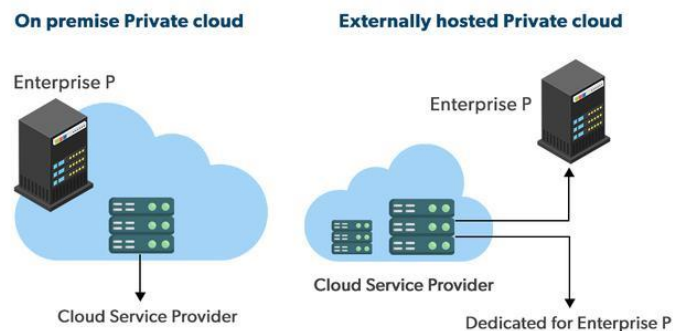
- **Minimal Investment:** Operates on a pay-per-use model with no substantial upfront capital expenditure.
- **Zero Setup & Maintenance:** The provider subsidizes and manages all physical hardware setup and ongoing maintenance.
- **Dynamic Scalability:** On-demand resources are nearly limitless, scaling instantly to fulfill business needs.

Disadvantages of the Public Cloud Model

- **Lower Security:** Because resources are shared publicly, meeting strict, high-level compliance and security standards can be challenging.
- **Low Customization:** The standardized environment offers little room for bespoke hardware or infrastructure customizations.

Private Cloud

The private cloud deployment model is the exact opposite of the public cloud deployment model. It's a one-on-one environment for a single user (customer). There is no need to share your hardware with anyone else. The distinction between [private and public clouds](#) is in how you handle all of the hardware. It is also called the "internal cloud" & it refers to the ability to access systems and services within a given border or organization. The cloud platform is implemented in a cloud-based secure environment that is protected by powerful firewalls and under the supervision of an organization's IT department. The private cloud gives greater flexibility of control over cloud resources.



Advantages of the Private Cloud Model

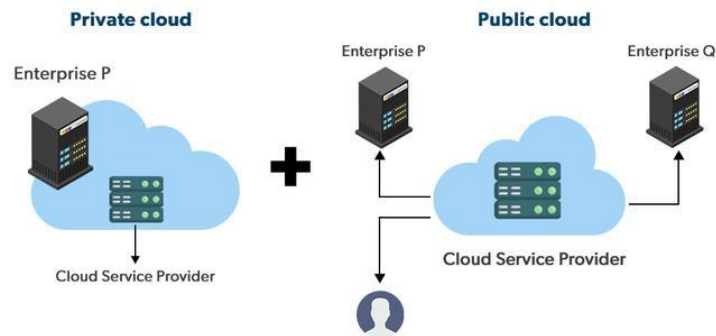
- **Total Control:** You have complete command over IT operations, service integration, and user behavior.
- **Elite Security and Privacy:** Ideal for sensitive corporate data, allowing for strict access controls and physical isolation.
- **Legacy System Support:** Highly customizable, making it capable of supporting legacy applications that cannot migrate to a public environment.

Disadvantages of the Private Cloud Model

- **High Cost:** Personalized facilities, dedicated hardware, and internal IT management make this the most expensive option.
- **Limited Scalability:** Scaling is restricted by the physical capacity of the dedicated hardware purchased.

Hybrid Cloud

By bridging the public and private worlds with a layer of proprietary software, hybrid cloud computing gives the best of both worlds. With a hybrid solution, you may host the app in a safe environment while taking advantage of the public cloud's cost savings. Organizations can move data and applications between different clouds using a combination of two or more cloud deployment methods, depending on their needs.



Advantages of the Hybrid Cloud Model

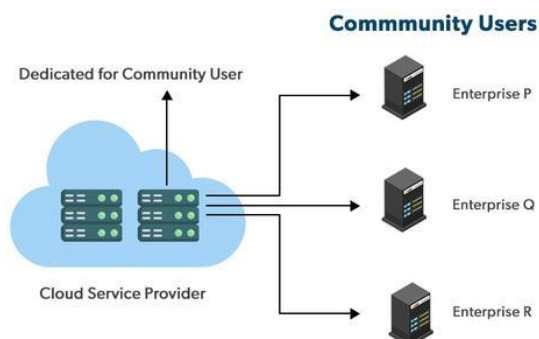
- **Ultimate Flexibility:** Businesses can keep sensitive data secure on the private cloud while running heavy, non-sensitive applications on the public cloud.
- **Cost Efficiency:** You only pay for additional public cloud capacity when you experience traffic spikes that your private cloud cannot handle (known as "cloud bursting").
- **Targeted Security:** Data theft risks are reduced by keeping the most critical assets completely isolated.

Disadvantages of the Hybrid Cloud Model

- **Difficult to manage:** Hybrid clouds are difficult to manage as it is a combination of both public and private cloud. So, it is complex.
- **Slow data transmission:** Data transmission in the hybrid cloud takes place through the public cloud so latency occurs.

Community Cloud

A community cloud is a collaborative infrastructure shared by a specific group of organizations from a similar industry that have shared concerns (e.g., compliance, security requirements, or mission goals). It can be managed internally by the organizations or by a third party.



Advantages of the Community Cloud Model

- **Cost Effective:** It is cost-effective because the cloud is shared by multiple organizations or communities.

- **Security:** Community cloud provides better security.
- **Shared resources:** It allows you to share resources, infrastructure, etc. with multiple organizations.
- **Collaboration and data sharing:** It is suitable for both collaboration and data sharing.

Disadvantages of the Community Cloud Model

- **Limited Scalability:** Community cloud is relatively less scalable as many organizations share the same resources according to their collaborative interests.
- **Rigid in customization:** As the data and resources are shared among different organizations according to their mutual interests if an organization wants some changes according to their needs they cannot do so because it will have an impact on other organizations.

How to Choose the Right Deployment Model

There is no one-size-fits-all approach. The ideal choice depends entirely on your specific business requirements. Consider these five factors:

- **Cost:** Determine your budget for upfront capital expenses versus ongoing operational expenses.
- **Scalability:** Assess your current traffic and predict how rapidly you will need to scale resources in the future.
- **Ease of Use:** Evaluate the technical expertise of your internal IT staff and their ability to manage complex infrastructures.
- **Compliance:** Identify any legal or industry regulations (like HIPAA or GDPR) that dictate where and how your data must be stored.
- **Privacy:** Classify your data to determine what can be public and what requires strict, isolated confidentiality.

Challenges and Risks in Cloud Adoption

Cloud adoption means moving an organization's applications, data, storage, and computing infrastructure from traditional/on-premises systems to cloud-based services.

Although cloud computing provides **scalability, flexibility, cost benefits, and easy access**, organizations may face several challenges and risks.

1. Security Risks

Sensitive data stored in the cloud may be exposed to:

- Unauthorized access
- Hacking and cyberattacks
- Malware and ransomware

- Weak authentication

Example: A company stores customer information in the cloud. If its cloud account credentials are stolen, attackers may access sensitive data.

2. Data Privacy

Organizations must ensure that personal and confidential information is properly protected.

Example: Hospitals storing patient records in the cloud must ensure that only authorized personnel can access them.

Risk: Improper handling of sensitive data can lead to privacy violations and legal problems.

3. Data Loss

Data may be accidentally deleted, corrupted, or affected by a security incident.

Solution: Organizations should maintain **regular backups and disaster-recovery mechanisms**.

4. Internet Dependency

Most cloud services require a reliable Internet connection.

If the Internet connection is slow or unavailable, users may not be able to access cloud applications or data.

Example: An online examination system hosted in the cloud may become inaccessible if the institution loses Internet connectivity.

5. Downtime and Service Availability

Cloud providers can experience:

- Hardware failures
- Network problems
- Software failures
- Maintenance-related outages

This can temporarily make applications unavailable.

6. Vendor Lock-in

Vendor lock-in occurs when an organization becomes highly dependent on one cloud provider and finds it difficult or expensive to move its applications and data to another provider.

Example: A company develops its entire application using one provider's proprietary cloud services. Moving to another provider may require significant modifications.

7. Compliance and Legal Issues

Organizations must comply with laws, regulations, and industry standards related to data storage and processing.

Example: A financial organization may have specific requirements regarding customer data and security.

8. Cost Management

Although cloud computing can reduce infrastructure costs, uncontrolled usage can result in unexpectedly high bills.

Example: A data science team continuously runs expensive GPU instances for machine-learning experiments without shutting them down.

Solution: Monitor resource usage and implement budgets and alerts.

9. Performance Issues

Cloud application performance can be affected by:

- Network latency
- Server load
- Bandwidth limitations
- Geographic distance from cloud resources

This can be particularly important for real-time applications.